1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

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1. Checking a Laptop Charger (Power Adapter)

Real-life Scenario:
A laptop does not turn on or charge even after being connected to the charger. The problem may be a faulty charger instead of the laptop itself.

A mustimeter is used to measure the output voltage of the laptop charger. If the measured voltage matches the value printed on the charger label (for example, **19V**), the charger is working properly. If the voltage is much lower or zero, the charger is faulty and should be replaced.

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A multimeter measures the charger's output voltage to verify whether it is supplying the correct power to the laptop.

2. Testing the CMOS Battery in a Desktop Computer

Real-life Scenario:
A desktop computer keeps showing the wrong date and time, loses BIOS settings after every shutdown, or displays a **CMOS Battery Failure** message during startup.

A multimeter checks the voltage of the CMOS battery (usually a **CR2032** battery). A healthy battery should read about **3 volts**. If the reading is much lower (such as **2.5V or less**), the battery is weak and should be replaced.

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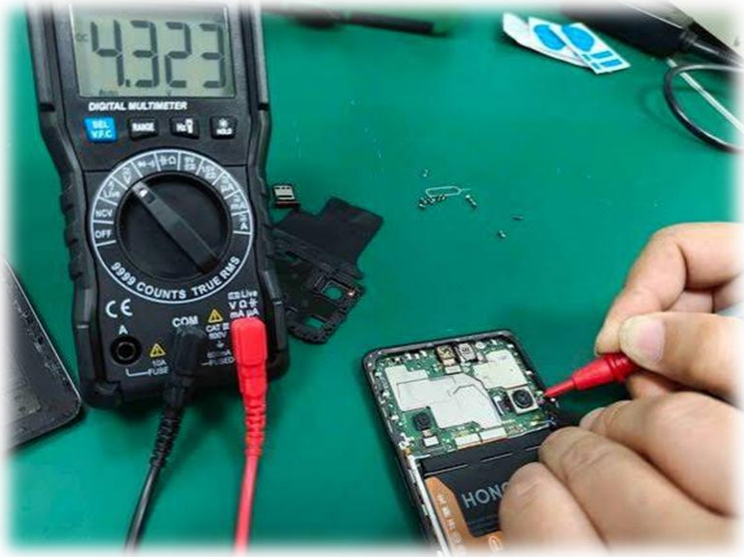
A multimeter checks the CMOS battery voltage to determine whether it is strong enough or needs replacement

3. Testing a Smartphone Battery

Real-life Scenario:
A smartphone does not turn on, shuts down unexpectedly, or does not charge properly even after using a working charger.



The multimeter measures the battery voltage to determine whether the battery has enough charge or has become faulty. If the voltage is very low or abnormal, the battery may need to be replaced.

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A multimeter measures the smartphone battery voltage to identify whether the battery is healthy, discharged, or defective.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video Hint: Focus on the codes shown and how they relate to hardware issues

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Summary:-

I watched a demonstration of a POST (Power-On Self-Test) card used to diagnose a motherboard problem. The card displayed hexadecimal POST codes that changed as the motherboard checked different hardware components during the boot process. When the code stopped at a particular value, it helped the technician determine where the hardware problem was occurring. POST cards are helpful for troubleshooting issues related to the motherboard, RAM, CPU, and BIOS.

YouTube Video: https://www.youtube.com/watch?v=RtzD6fZ532k

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details Constraint: Blur or crop out any personal information before submitting.

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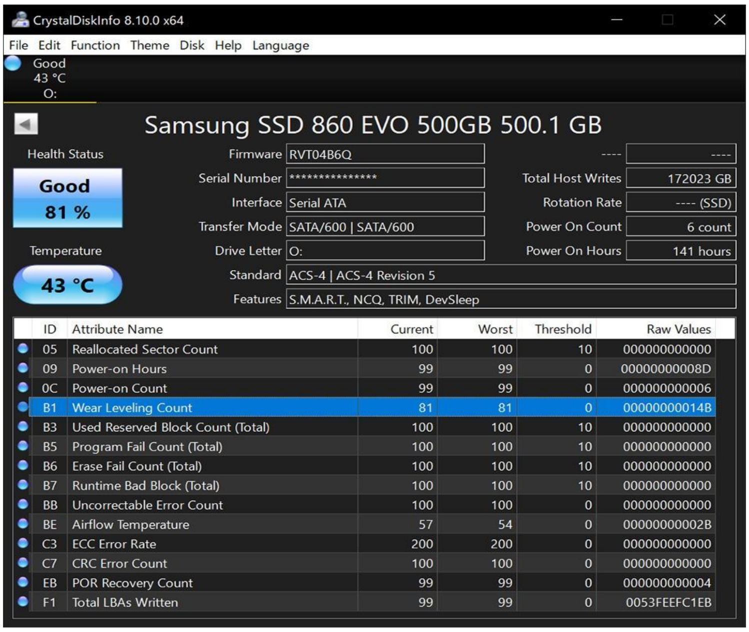
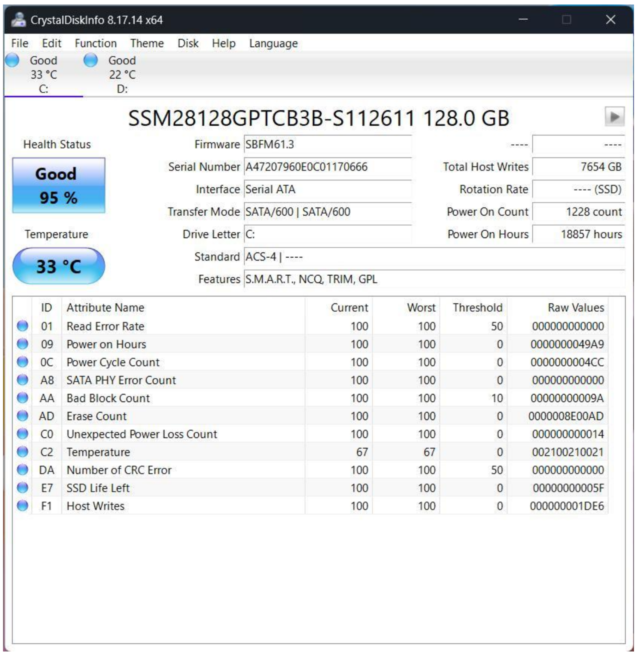
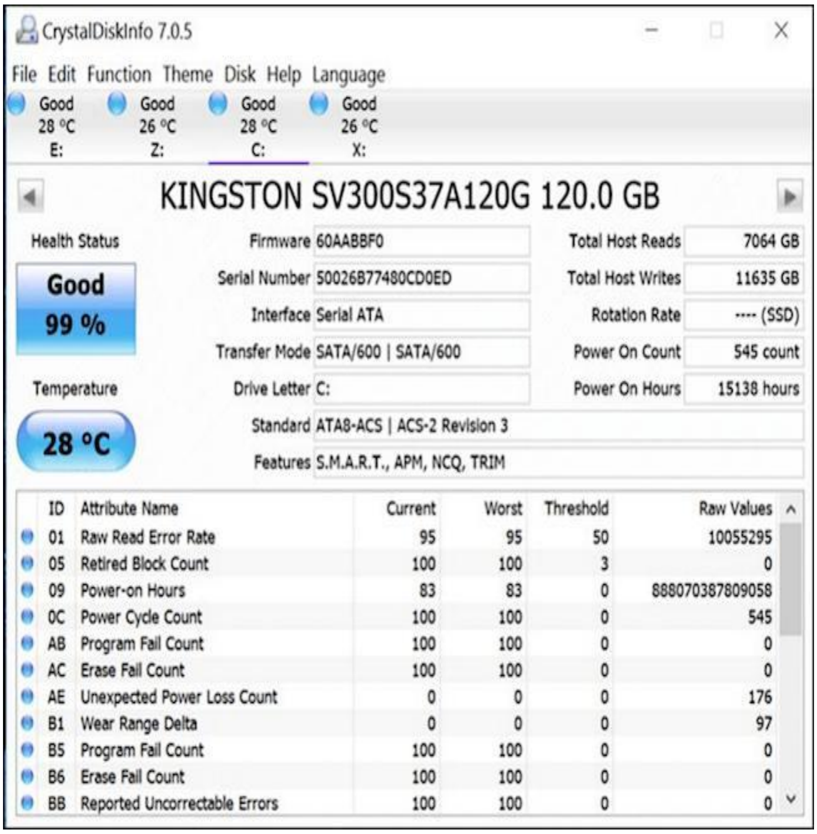
Diagnostic Tool – CrystalDiskInfo

I downloaded and installed CrystalDiskInfo, a free diagnostic tool, on my laptop. I opened the software and ran a diagnostic check on my storage drive. The main dashboard displayed the drive's health status, temperature, model information, and S.M.A.R.T. details. The health status showed that the storage drive was working normally. I took a screenshot of the main dashboard and cropped/blurred any personal information before submitting it.

Screenshot:







4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	**What it checks**	**Example Problem Detected**
Multimeter	**Checks electrical values such as voltage, resistance, and continuity.**	**Detects a faulty power supply or incorrect voltage on a motherboard.**
POST Card	**Checks the motherboard's Power-On Self-Test process using POST/BIOS codes.**	**Detects a problem with RAM, CPU, BIOS, or motherboard during startup.**

Tool	What it checks	Example Problem Detected
Software Diagnostics	Checks the health and performance of hardware such as HDD/SSD, RAM, CPU, and temperatures.	Detects a failing hard drive/SSD or overheating component.